

HONGYI YANG

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EDUCATION

Zhejiang University

Sep 2023 - Jun 2027 (expected)

B.Eng. in Optoelectronic Information Science and Engineering; Chu Kochen Honors College (Mixed Class)

- **GPA:** 4.11 / 4.30 **Rank:** 4 / 162 (Chu Kochen Honors College); 4 / 102 (School of Optoelectronic Science and Engineering)
- **TOEFL:** 103 (Reading 27 / Listening 27 / Speaking 25 / Writing 24)

PUBLICATIONS & MANUSCRIPTS

† equal contribution | * corresponding author

- **Hongyi Yang**†, Haipeng Zhu†, Shiyu Feng, Zexi Lu, Xiyao Peng, Hongsheng Chen*, Sihan Zhao*, Dexin Ye*, and Haoliang Qian*. "Sub-Poissonian Electroluminescence from Resonant Inelastic Tunneling in One-Dimensional Gold Atomic Chains." *ACS Photonics*, published online October 2, 2026. [DOI: [10.1021/acsp Photonics.6c01563](https://doi.org/10.1021/acsp Photonics.6c01563)]
- **Hongyi Yang**, Wendi Xia, Jun Li, Dexin Ye*, and Haoliang Qian*. "Physically Grounded Neural Networks: From Physical Substrates to Physics-Integrated Learning." *Applied Physics Reviews*, 2026 (under review).

RESEARCH EXPERIENCE

Xu Research Group, Stanford University

Stanford, USA

Visiting Undergraduate Research Intern; Advisor: Prof. Sheng Xu

Jul - Oct 2026 (on-site)

- **Strain-Induced Ferroelectricity in FAPbI₃**
 - Built DFT models of epitaxially strained FAPbI₃ in Quantum ESPRESSO, including phonon calculations of the strained structures.
 - Mapped the off-centre Pb displacement and its double-well energy landscape; found that the barrier deepens under compressive strain.
 - Related the polarization along z to Pb off-centring and identified the two wells as the origin of switchable ferroelectric states.

State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University

Hangzhou, China

Student Researcher; Advisor: Prof. Haoliang Qian

Apr 2025 - present

- **Electrically Driven Non-Classical Light from Gold Atomic Chains**
 - Built a quantum-transport model of a gold atomic-chain junction from relativistic DFT, Wannier functions and DFPT vibrational modes, with bulk lead self-energies.
 - Solved the non-equilibrium Green's function problem without the wide-band approximation and treated electron-phonon coupling in the self-consistent Born approximation; implemented the pipeline in Python on HPC clusters.
 - Calculated current and current noise to identify sub-Poissonian electron statistics and studied how scattering and substitutional doping alter photon statistics.
 - Coupled the transport model to full-wave simulations of a plasmonic nanocube dimer to connect resonant inelastic tunneling with emitted light.

International Research Center for Advanced Photonics, Zhejiang University

Hangzhou, China

Research Assistant; Advisor: Prof. Dawei Di

Sep 2023 - Jun 2024

- **Perovskite Optoelectronic Devices**
 - Learned cleanroom processes for perovskite devices, including spin-coating and encapsulation.
 - Designed and built the optical setup used to measure external quantum efficiency.

SCHOLARSHIPS

- 2025-2026** Huawei Elite Scholarship
- 2024-2025** OPPO 'Benfen' Scholarship (8 recipients nationwide per year)
- 2023-2024** National Scholarship (top 0.2% nationwide)
- 2023-2024** First-Class Scholarship of Zhejiang University; Excellent Student Title

HONORS & AWARDS

- 2025-2026** First Prize, National Undergraduate Optoelectronic Design Competition
- Oct 2025** Top Ten Undergraduates, School of Optoelectronic Science and Engineering, Zhejiang University
- 2024** First Prize, Zhejiang Provincial College Physics Competition
- 2023-2025** 3rd Place (Long Jump), 5th Place (Triple Jump), Zhejiang University Sports Meeting
- 2022** First Prize, National High School Physics Olympiad (Hunan Province)

TECHNICAL SKILLS

Programming: Python, PyTorch, C/C++, LaTeX; high-performance computing (SLURM clusters)

Simulation: custom NEGF transport codes, Quantum ESPRESSO (DFT/DFPT), Wannier90, COMSOL Multiphysics, Zemax

Hardware: SolidWorks, Altium Designer

RESEARCH INTERESTS

Quantum photonics: integrated quantum photonics and on-chip quantum light sources

Device modelling: first-principles modelling of optoelectronic devices (NEGF, DFT/DFPT, Wannier functions)

Integration: hybrid integration of low-dimensional materials on silicon photonic platforms